WORKED SOLUTION

STEP III Specimen paper, Question 8

Pure • Vectors and Cauchy-Schwarz

STEP III Specimen paper, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s3-q8

The key idea

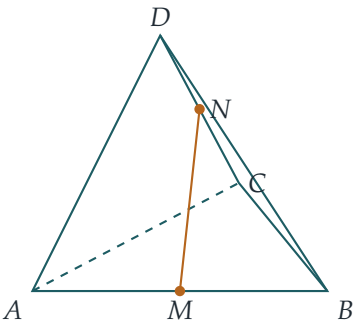
In part (i), give the four vertices position vectors. Every length we need is the modulus of a simple combination of them, and squared lengths are dot products we can expand. Both sides then come out as the same expression, and symmetry does most of the bookkeeping. In part (ii), the left-hand side looks like a dot product with the fixed vector $(3, 2, 6)$, whose length is a whole number. That is exactly what the Cauchy–Schwarz inequality is made for.

Solution

Part (i)

Let the tetrahedron be $ABCD$, with position vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d}$ relative to any origin. Its six edges come in three pairs of opposite edges (edges with no common vertex):

$$AB \text{ and } CD, \qquad AC \text{ and } BD, \qquad AD \text{ and } BC.$$



For the pair AB, CD: the midpoints are $M = \frac{1}{2}(\mathbf{a} + \mathbf{b})$ and $N = \frac{1}{2}(\mathbf{c} + \mathbf{d})$, so

$$MN^2 = \tfrac{1}{4}\,|\mathbf{c} + \mathbf{d} - \mathbf{a} - \mathbf{b}|^2.$$

The other two midlines are found in the same way, so four times the sum of the squares of the three midlines is

$$S = |-\mathbf{a}-\mathbf{b}+\mathbf{c}+\mathbf{d}|^2 + |-\mathbf{a}+\mathbf{b}-\mathbf{c}+\mathbf{d}|^2 + |-\mathbf{a}+\mathbf{b}+\mathbf{c}-\mathbf{d}|^2.$$

Expand each term using $|\mathbf{u}|^2 = \mathbf{u}\cdot\mathbf{u}$. Each term contributes $|\mathbf{a}|^2+|\mathbf{b}|^2+|\mathbf{c}|^2+|\mathbf{d}|^2$, plus cross terms $\pm 2\,\mathbf{u}\cdot\mathbf{v}$, with a plus sign when $\mathbf{u}$ and $\mathbf{v}$ have the same sign in that term.

Ask yourself How do the cross terms add up across the three terms?

Take any pair of vertices, say $\mathbf{a}$ and $\mathbf{b}$. They have the same sign in the first term and opposite signs in the other two, so their cross terms total $2\,\mathbf{a}\cdot\mathbf{b} - 2\,\mathbf{a}\cdot\mathbf{b} - 2\,\mathbf{a}\cdot\mathbf{b} = -2\,\mathbf{a}\cdot\mathbf{b}$. The same holds for every pair: each of the three terms splits the four vertices into two pairs (a vertex pair together with its opposite pair), and every vertex pair has matching signs in exactly one of the three terms. Hence

$$S = 3(|\mathbf{a}|^2+|\mathbf{b}|^2+|\mathbf{c}|^2+|\mathbf{d}|^2) - 2(\mathbf{a}\cdot\mathbf{b}+\mathbf{a}\cdot\mathbf{c}+\mathbf{a}\cdot\mathbf{d}+\mathbf{b}\cdot\mathbf{c}+\mathbf{b}\cdot\mathbf{d}+\mathbf{c}\cdot\mathbf{d}).$$

Now the edges. For example, $AB^2 = |\mathbf{b}-\mathbf{a}|^2 = |\mathbf{a}|^2+|\mathbf{b}|^2-2\,\mathbf{a}\cdot\mathbf{b}$. Summing over all six edges, each vertex lies on three edges, so each $|\mathbf{a}|^2$ appears three times, and each pair gives one $-2\,\mathbf{u}\cdot\mathbf{v}$:

$$AB^2+AC^2+AD^2+BC^2+BD^2+CD^2 = 3\sum|\mathbf{a}|^2 - 2\sum_{\text{pairs}}\mathbf{a}\cdot\mathbf{b} = S.$$

So the sum of the squares of the edges equals four times the sum of the squares of the three lines joining midpoints of opposite edges, as required.

Part (ii)

Take

$$\mathbf{a} = \begin{pmatrix}3\\2\\6\end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix}x\\y\\1\end{pmatrix}, \qquad \text{so} \qquad \mathbf{a}\cdot\mathbf{b} = 3x+2y+6,$$

with $|\mathbf{a}| = \sqrt{9+4+36} = 7$ and $|\mathbf{b}| = \sqrt{x^2+y^2+1}$. Since any real number is at most its modulus,

$$3x+2y+6 = \mathbf{a}\cdot\mathbf{b} \le |\mathbf{a}\cdot\mathbf{b}| \le |\mathbf{a}|\,|\mathbf{b}| = 7\sqrt{x^2+y^2+1},$$

which is the required inequality.

Equality. We need equality at both steps.

- $|\mathbf{a}\cdot\mathbf{b}| = |\mathbf{a}||\mathbf{b}|$ holds exactly when $\mathbf{a}$ and $\mathbf{b}$ are parallel, since $\mathbf{a}\cdot\mathbf{b} = |\mathbf{a}||\mathbf{b}|\cos\theta$ and we need $|\cos\theta| = 1$. As $\mathbf{a} \neq \mathbf{0}$, this means $\mathbf{b} = \lambda\mathbf{a}$ for some λ. The third components give $1 = 6\lambda$, so $\lambda = \frac{1}{6}$ and $x = \frac{1}{2}$, $y = \frac{1}{3}$.
- $\mathbf{a}\cdot\mathbf{b} = |\mathbf{a}\cdot\mathbf{b}|$ needs $\mathbf{a}\cdot\mathbf{b} \ge 0$. Here $\mathbf{a}\cdot\mathbf{b} = \lambda|\mathbf{a}|^2 = \frac{49}{6} > 0$, so this holds too.

Directly: at $x = \frac{1}{2}$, $y = \frac{1}{3}$ the left side is $\frac{3}{2} + \frac{2}{3} + 6 = \frac{49}{6}$ and the right side is $7\sqrt{\frac{1}{4} + \frac{1}{9} + 1} = 7 \cdot \frac{7}{6} = \frac{49}{6}$. So equality holds if and only if

$$x = \tfrac{1}{2}, \qquad y = \tfrac{1}{3}.$$

Checks

- For a regular tetrahedron with edge 1, the six squared edges sum to 6. Placing it at alternate corners of a cube of side $\frac{1}{\sqrt{2}}$, each midline joins centres of opposite faces, so it has length $\frac{1}{\sqrt{2}}$, and $4 \times 3 \times \frac{1}{2} = 6$.
- A numerical test on 10^4 random tetrahedra agreed with the identity to rounding error. For (ii), $7\sqrt{x^2 + y^2 + 1} - (3x + 2y + 6)$ was positive at 2×10^5 random points, and its smallest value on a fine grid was at $(0.5, 0.335)$, close to $(\frac{1}{2}, \frac{1}{3})$, where it is 0.

Common slips

- Forgetting the step $\mathbf{a} \cdot \mathbf{b} \leq |\mathbf{a} \cdot \mathbf{b}|$, or ignoring it in the equality case. Here it causes no trouble because $\lambda > 0$, but it has to be checked: with $\lambda < 0$ the two sides would differ in sign.
- Using $(3, 2, 6)$ with $(x, y, 1)$ but taking $|\mathbf{a}|$ to be anything other than 7. The question is built so that $3^2 + 2^2 + 6^2$ is a perfect square.

Beyond STEP

Part (i) is a three-dimensional relative of the parallelogram law $|\mathbf{u} + \mathbf{v}|^2 + |\mathbf{u} - \mathbf{v}|^2 = 2|\mathbf{u}|^2 + 2|\mathbf{v}|^2$. In fact the three midlines meet at the centroid and bisect each other. Part (ii) is the Cauchy–Schwarz inequality, which holds in any inner-product space, including spaces of functions, where it bounds integrals. Strang's *Introduction to Linear Algebra* is a good place to see both ideas in a wider setting.

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